



VGH-TP VPC workflow

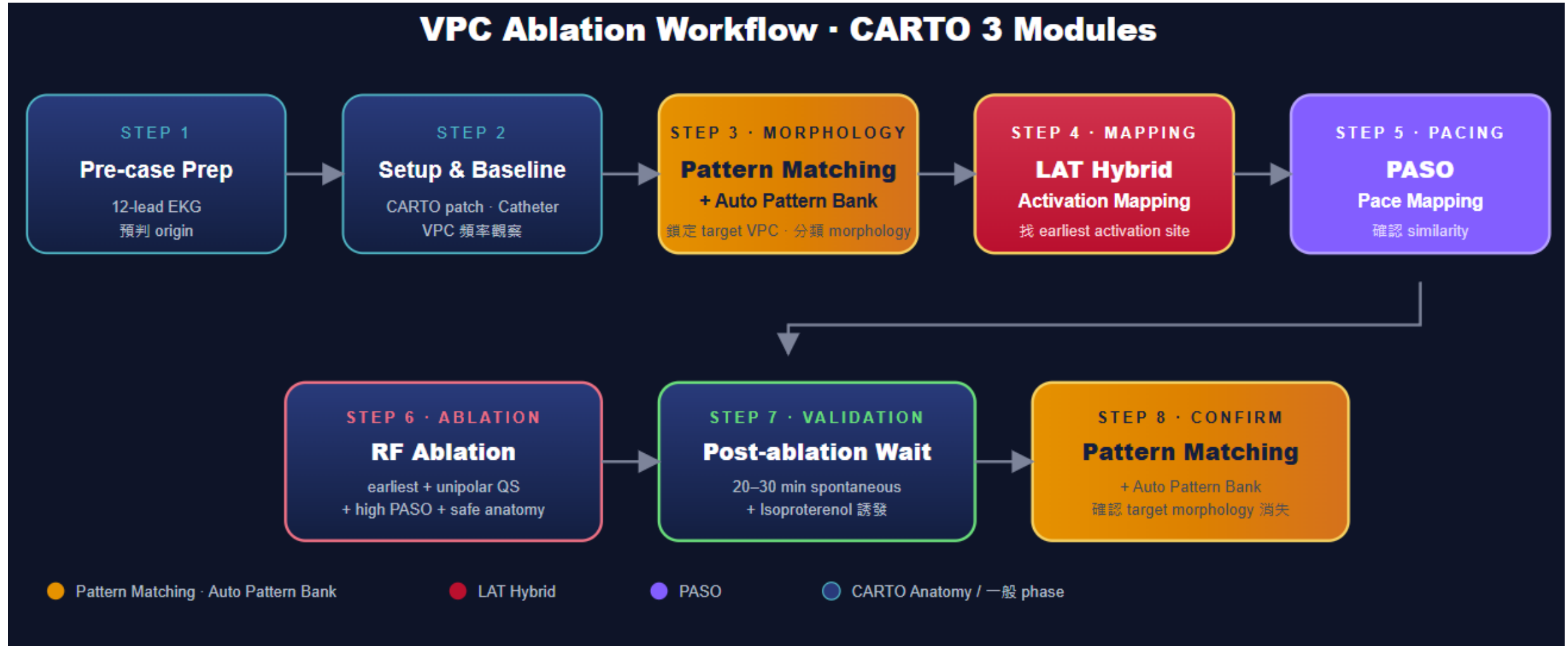
Jason



Step 1 Pre-case Prep

- 備機前確認
 - 導管室
 - 主診斷 (AF, VPC, PSVT.....)
 - 主治醫師 (手術習慣、常用導管類型)
 - 術式 (導管與 ablation 機器)
 - 是否研究案? (IIS, CL, DE)

VPC ablation workflow overview



Step 1 Pre-case Prep

- 備機

- PIU
- Workstation
- SMA + remote
- Patch
- Location pad + 支架

- 線材

- 導管
- ECG in
- ECG out to recording
- Cable
- Indifferent Electrode

Step 2 Setup & Baseline

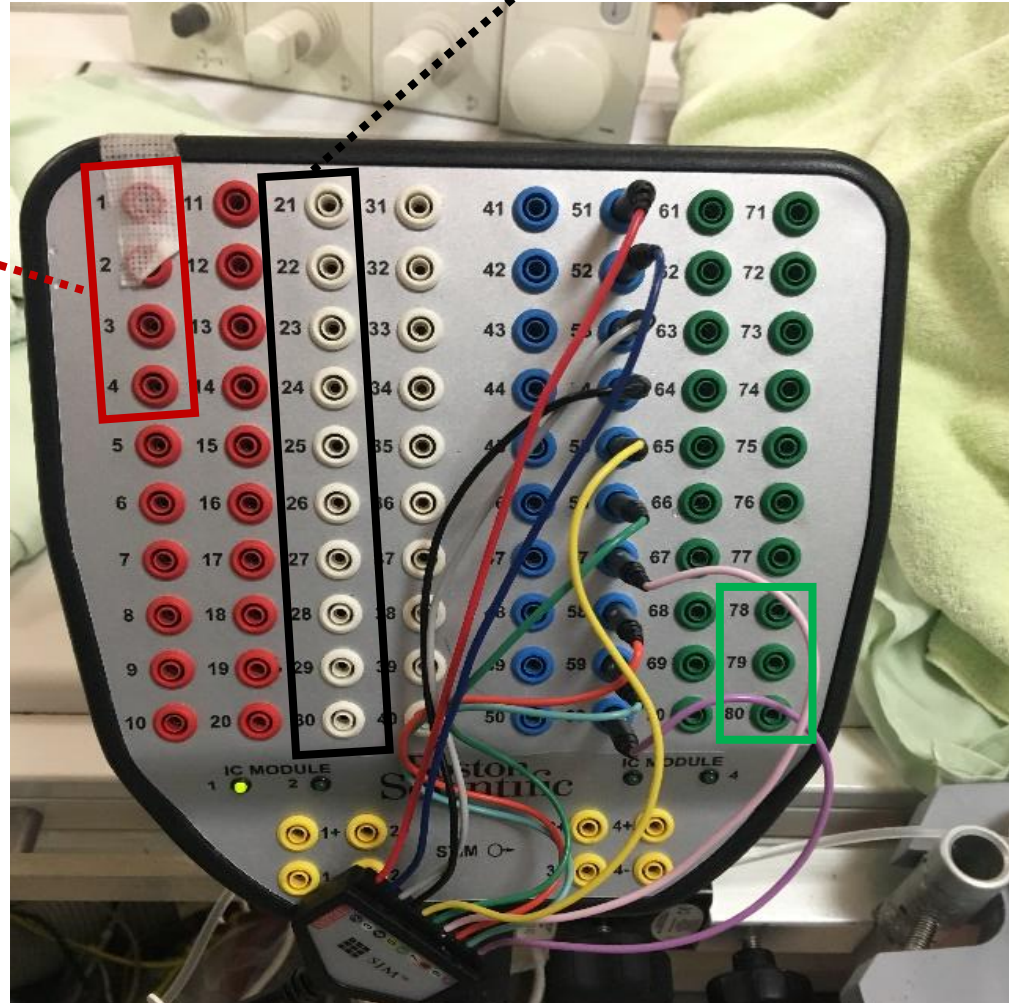
- Catheter 配置
 - 將當日所需的導管放置於control room 外窗邊
- 連接cable 至PIU
 - 將沒有忌諱要先接導管的cable連接至PIU
 - 並將cable 統一掛好於台車上 (方便護理師套無菌套)



IC out (彩虹線) 配置

Pin 21-30
20pA (DECA)

Pin 1-4
MAP (TC)



Step 2 Setup & Baseline

1

導管室

確認 lab 安排、
設備可用、有
沒有撞時段

2

主診斷

AF / VPC /
PSVT / VT 決
定 mapping 策
略

3

主治醫師

手術習慣、常用
導管類型、偏好
的 setup

4

術式

導管與 ablation
機器

5

研究案？

IIS · CL · DE —
影響
documentation
與耗材

Step 2 Setup & Baseline

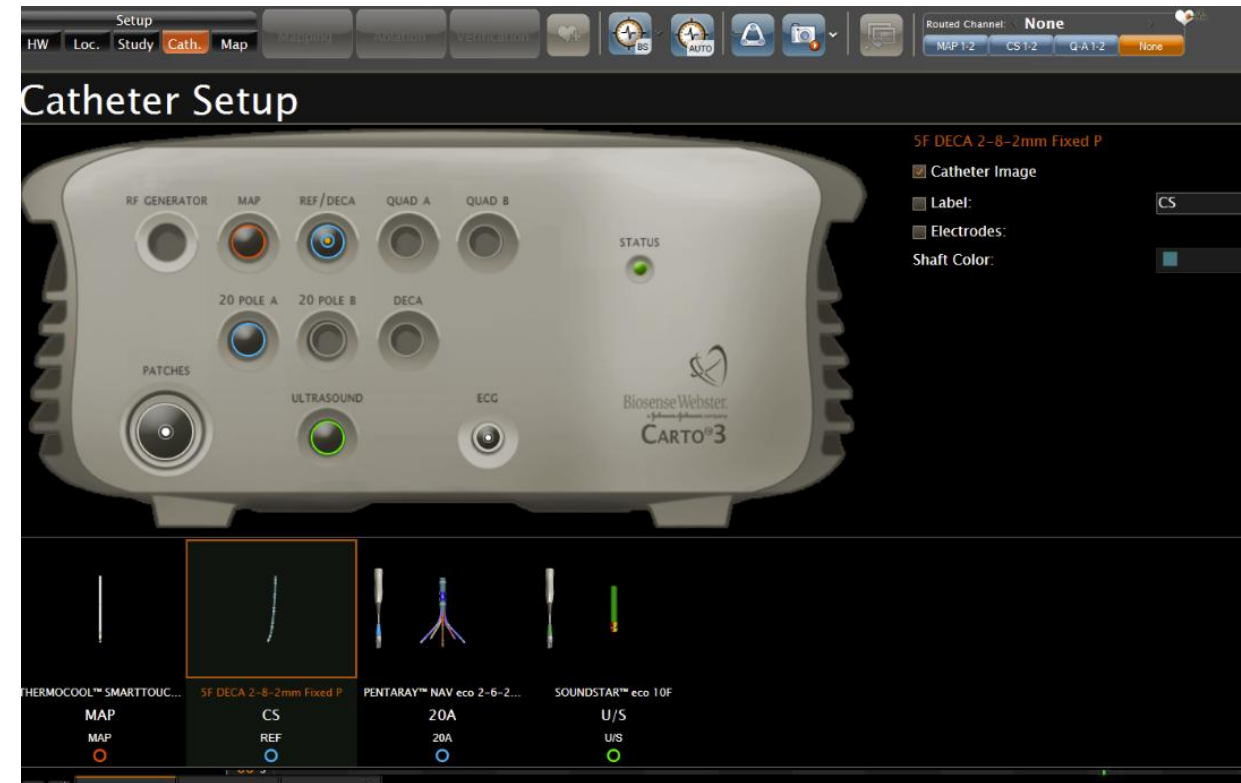
- **CARTO case setup**
 - Study setup: patient name, ID, physician, select template

The screenshot displays the 'Study Setup' window in the CARTO3 software. The window is organized into four main sections:

- Patient Details:** Includes input fields for 'Last Name', 'First Name', 'ID', and 'Date of Birth' (with dropdowns for month and year). A 'Gender' section has radio buttons for 'Male' and 'Female'. A 'Patient List...' button is also present.
- Additional Details:** Includes a 'Comments' text area, a 'Study Name' text field, and a 'Physician' dropdown menu.
- Template:** A list of arrhythmia templates: 'Atrial Fibrillation', 'Atrial Flutter', 'Focal Atrial Tachycardia', 'Accessory Pathway', 'AVNRT', 'Ventricular Tachycardia', and 'Other'.
- Connected Machines:** Includes 'Recording' status (set to 'No communication'), 'Ultrasound' (set to '<No Ultrasound>'), and 'Generators' (set to 'No Generators').

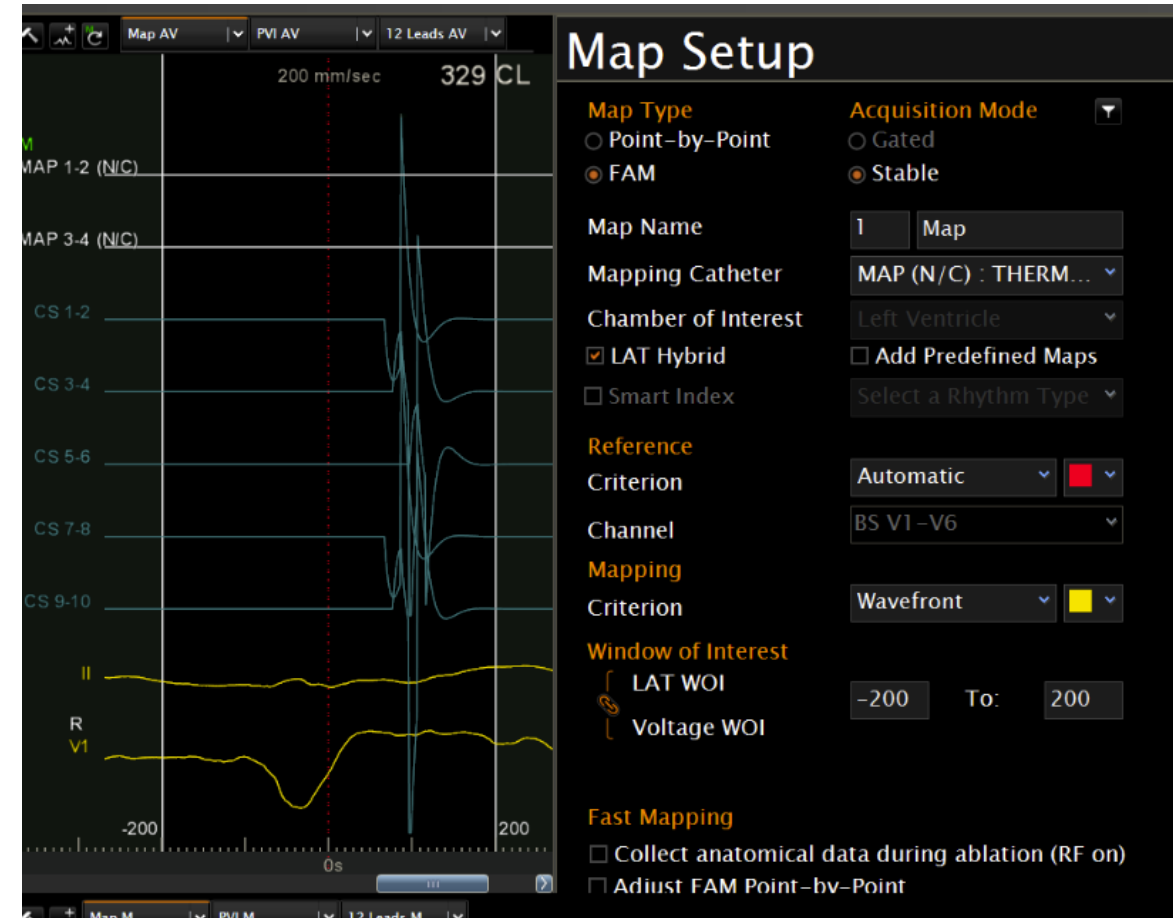
Step 2 Setup & Baseline

- Catheter setup
 - CS: DECANAV
 - Ablation tube: THERMOCOOL



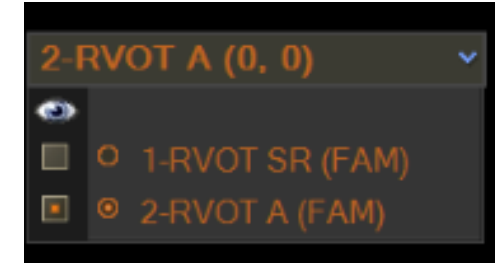
Step 2 Setup & Baseline

- Map setup
 - 先建好幾個可能會用到的圖 (RVOT SR, RVOT A, CS A, CUSP A, RVOT B)
 - Chamber 選擇 Ventricular，勾選LAT hybrid
 - Reference 使用 BS V1-V6 Automatic
 - WOI 預調 -180 to -20



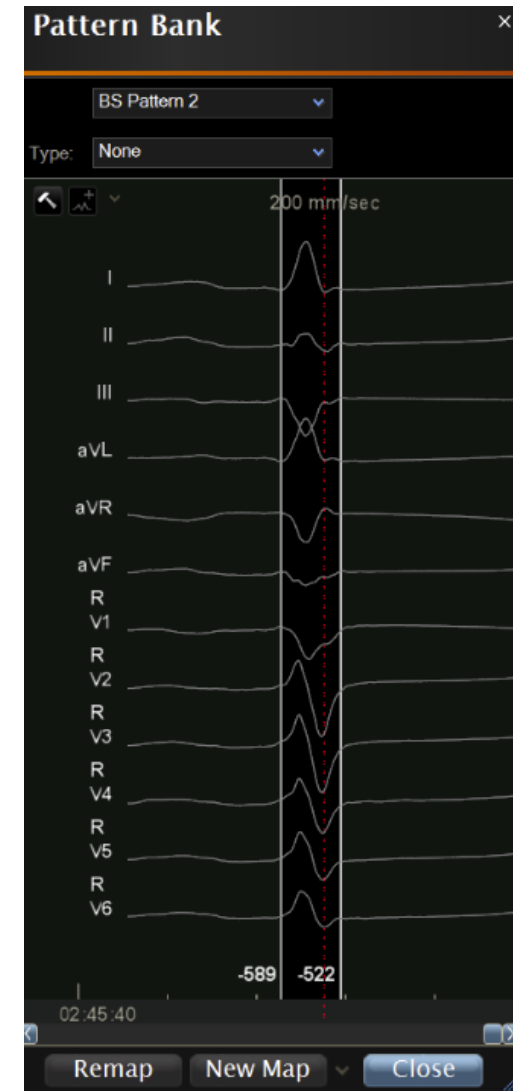
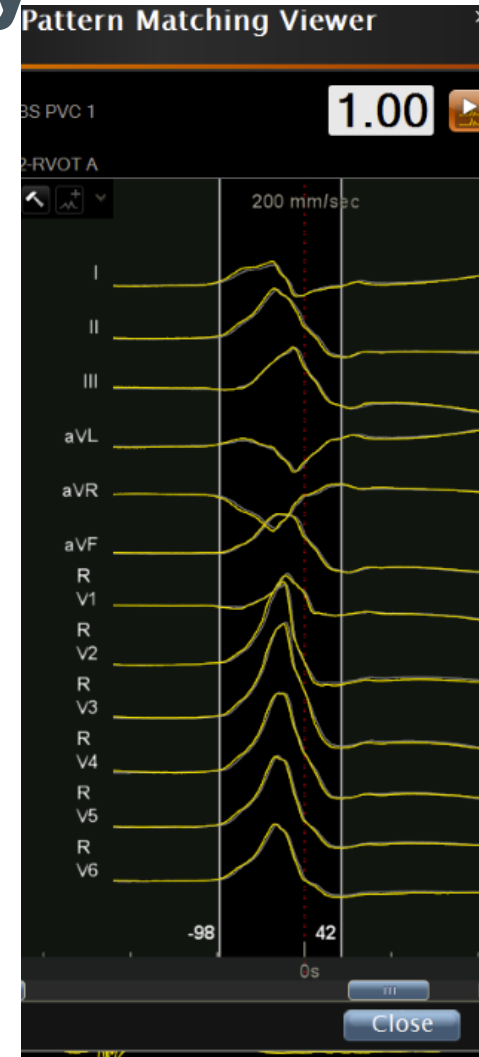
Step 2 Setup & Baseline

- Mapping 頁面
 - 畫面配置: 主畫面左邊 & 副畫面右邊，並開啟 PASO view
 - 將預先設定好的Map 進行parallel mapping (最多可以設四張Map)



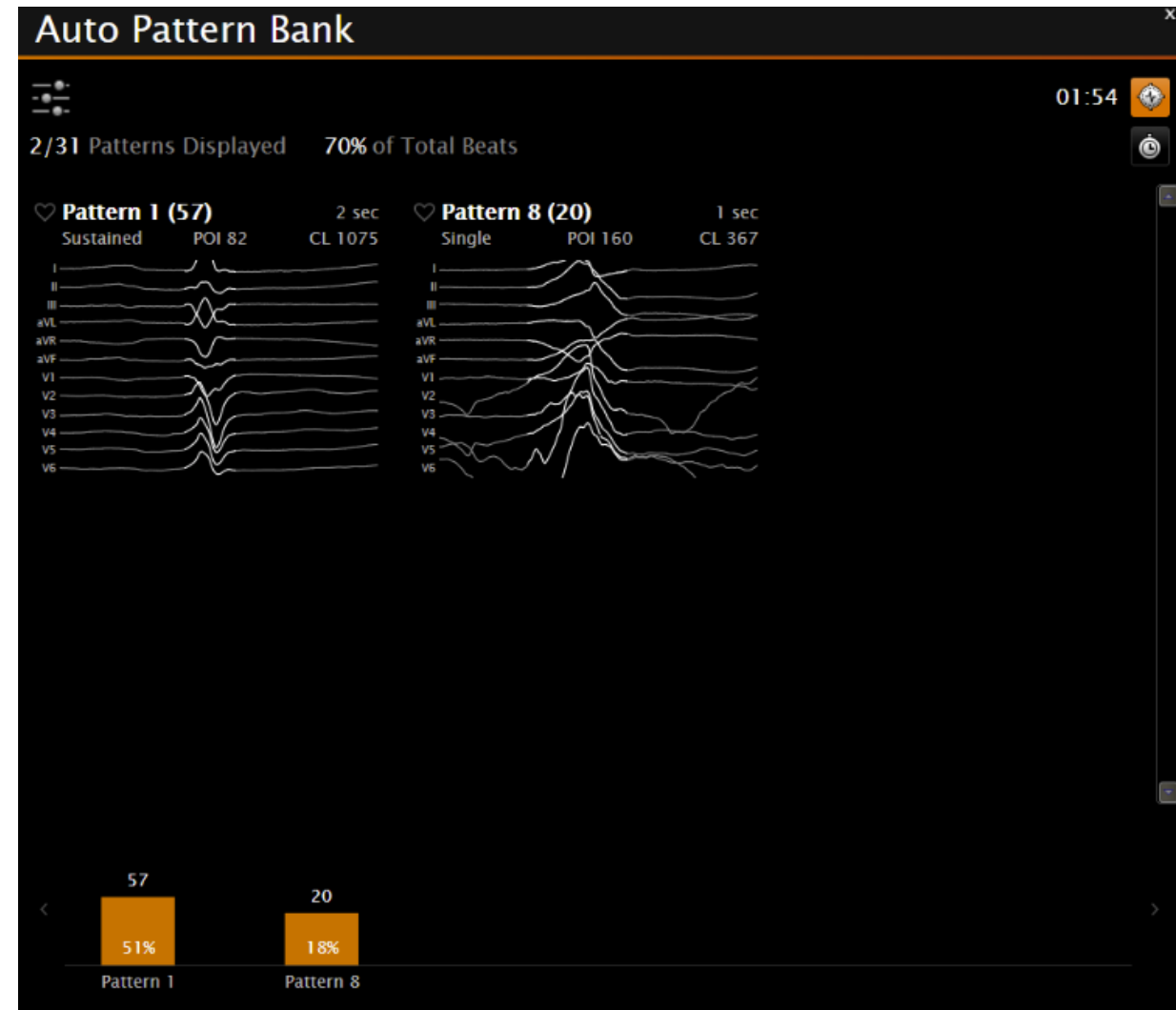
Step 3 Target VPC morphology

- 12 lead ECG 預判
- Template 抓取
 - 選取選單上方的時鐘，使用 pattern matching 並抓取 SR, VPC(可能數個) template
 - 各個Map confidence 設定好pattern matching template, Do not require while pacing 打勾



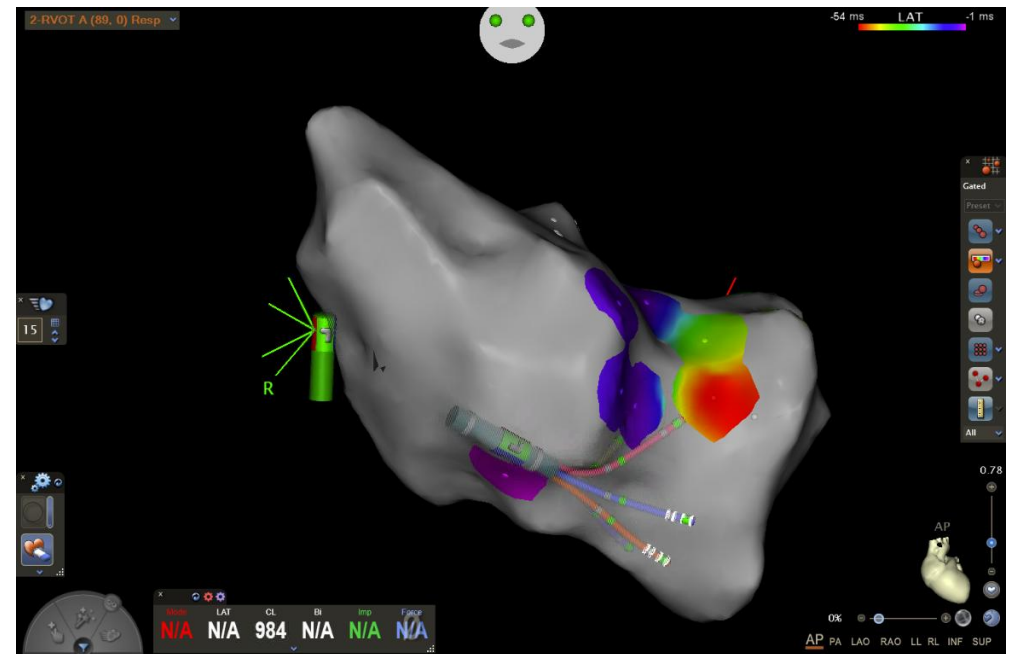
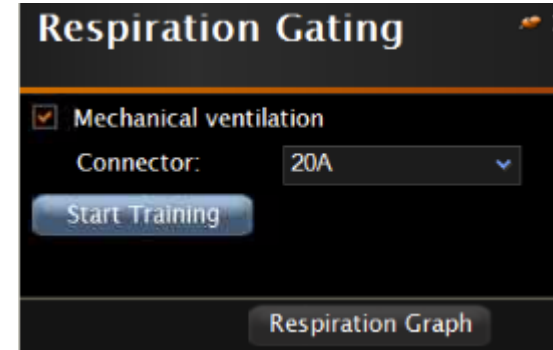
Step 3 Target VPC morphology

- 開啟 Auto pattern bank
 - 收取各 template beat counts



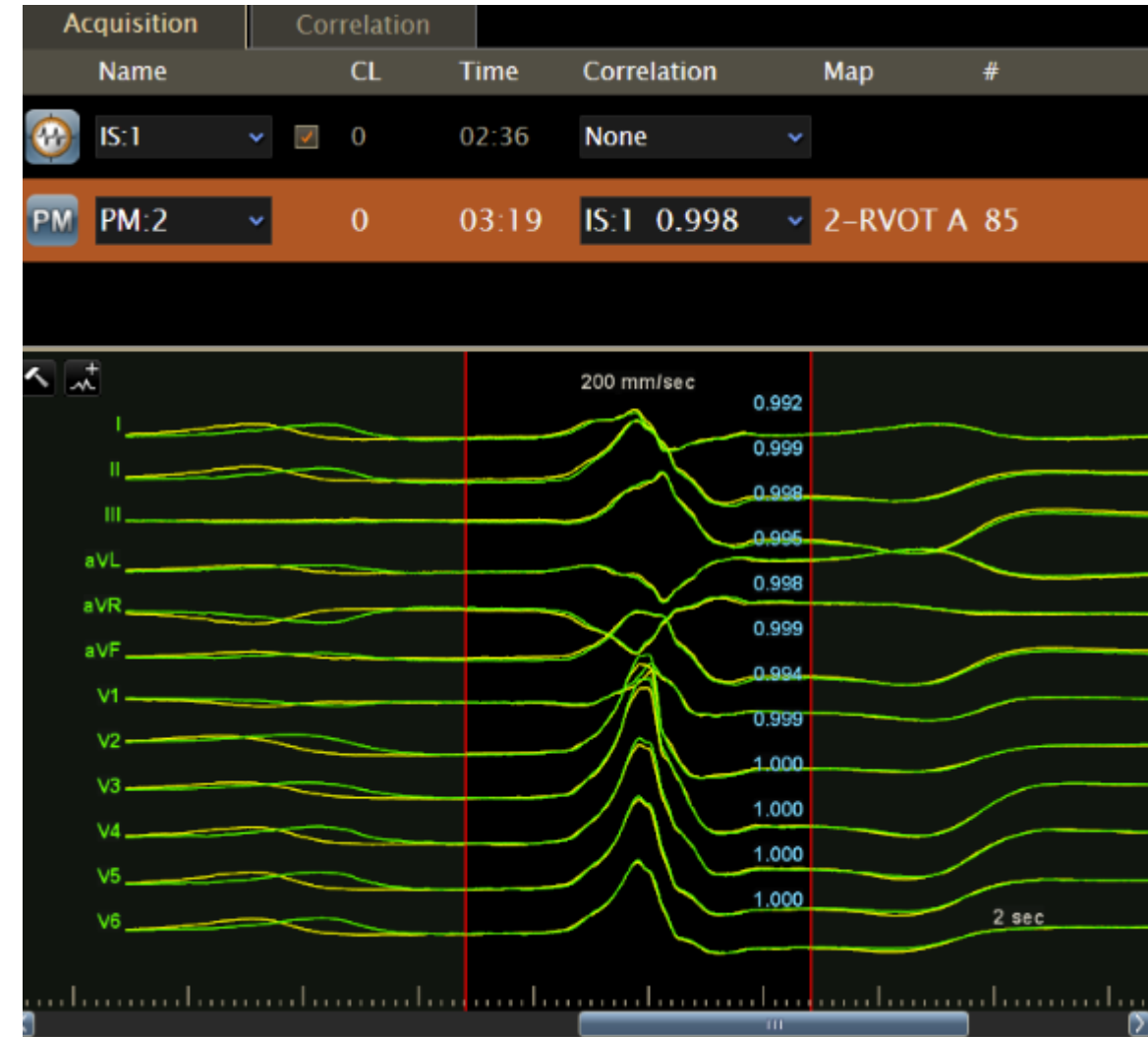
Step 4 Mapping

- 開啟 ACCURESP
 - Enable respiration gating
 - 導管進入後要找地方收呼吸
- 開始用DECANAV 收FAM & EA



Step 4 Mapping

- **PASO 確認相似度**
 - 以臨床 12-lead VT 波形為 template
 - Pace mapping score >90% 視為接近 exit site



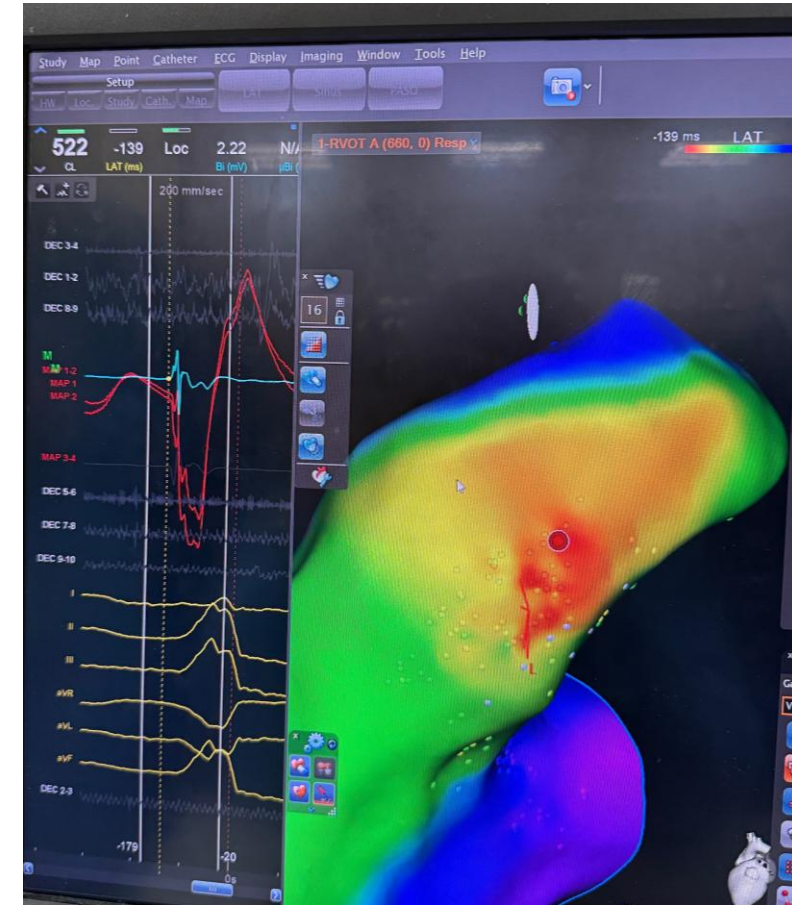
Step 5 Ablation

- **Target Selection Checklist**

- 不是 fusion beat
- LAT timing 足夠早
- local EGM 清楚，不是 far-field
- unipolar 呈 QS 或接近 QS
- PASO score 夠高
- catheter contact 穩定
- anatomy 位置合理
- 遠離 His / coronary artery / valve / phrenic nerve
- 若在 cusp / summit，已考慮 coronary distance



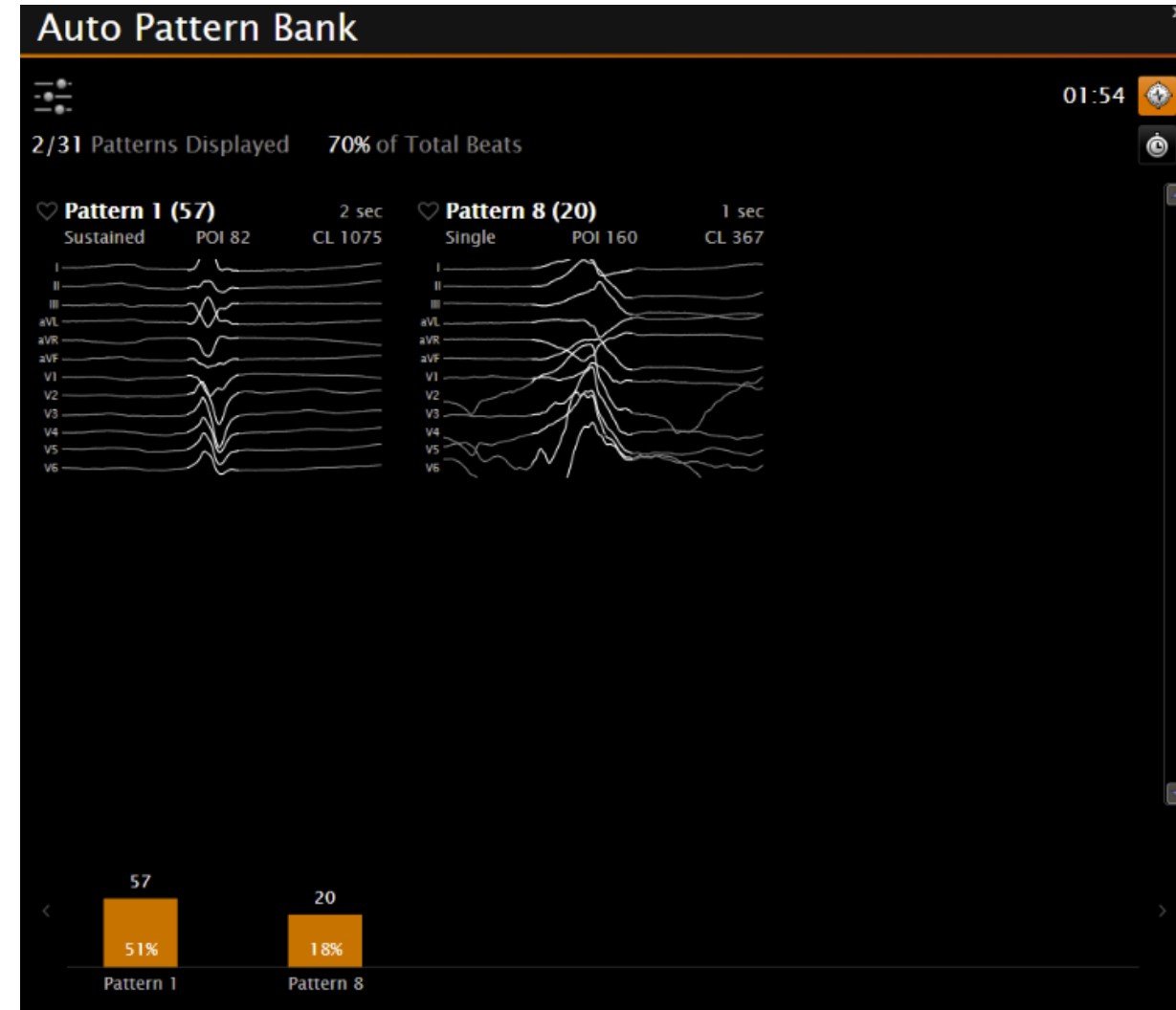
Pic1. 3D Map with Corr



Pic2. 3D Map with LAT

Step 5 Ablation

- **Post ablation validation**
 - 等待至少 20–30 分鐘觀察 Spontaneous PVC
 - 誘發 (Isoproterenol, Burst pacing)
- **Auto Pattern Bank 確認**





Thank you

